Exercise Sheet 12

Conventions follow Lecture 12. Haar measures have total mass 1, $(L_g f)(x) = f(g^{-1}x)$, $(R_g f)(x) = f(xg)$, and $c_{u,v}(x) = \langle u, \Pi(x)v\rangle$ with the inner product antilinear in the first slot. "The multiplicity theorem" means: σ occurs in $L^2(K/H)$ exactly $\dim V_\sigma^H$ times. "Reciprocity" means $m_\sigma(L^2\text{-}\operatorname{Ind}_H^K W) = \dim \operatorname{Hom}_H(V_\sigma|_H, W)$, with the maps $\Phi(T)v = (Tv)(e)$ and $(\Psi(A)v)(x) = A(\Pi(x)^{-1}v)$. Weights for a circle group $\{R_3(\theta)\}$ are defined as in Lecture 12. In SO(3) contexts, an unqualified V_ℓ is the SO(3)-module of dimension $2\ell+1$. The SU(2)-modules V_m of Lecture 8 are always named as such. Exercise 12.5 is the optional extension.

Exercise 12.1 (Labels on the circle). Let $K = \mathrm{U}(1) = \{e^{i\theta}\}$ with Haar measure $d\theta/2\pi$. For $n \in \mathbb{Z}$ let $\chi_n(e^{i\theta}) = e^{in\theta}$ act on $\mathbb{C}$ with the standard inner product. Fix $N \geq 1$ and let $H = \mu_N = \{e^{2\pi ik/N} : 0 \leq k < N\}$.

(a) Show that $E_{\chi_n} = \mathbb{C}\, e^{in\theta}$. Compute $L_{e^{i\varphi}} e^{in\theta}$. Which character occurs in E_{χ_n}? Compare with $E_\sigma \cong \overline{V_\sigma} \otimes V_\sigma$.

(b) Compute $\dim \chi_n^H$. Use the multiplicity theorem to list the characters in $L^2(K/H)$, with multiplicities.

(c) Check (b) directly. Which functions $e^{in\theta}$ are right-H-invariant?

Exercise 12.2 (Induction from the torus of SU(2)). Let $K = \mathrm{SU}(2)$ and $T = \{t_z\}$, where

$$t_z = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}, \qquad |z| = 1.$$

For $k \in \mathbb{Z}$ let $\chi_k(t_z) = z^k$. Use the following facts from Lecture 8. The irreducibles of SU(2) are V_m, $m \geq 0$, of dimension $m+1$ and spin $m/2$. Each V_m has a basis $v_0, \ldots, v_m$ with $t_z v_a = z^{m-2a} v_a$ for all $|z| = 1$.

(a) Compute $\dim \operatorname{Hom}_T(V_m, \chi_k)$.

(b) Find the multiplicity of V_m in $L^2\text{-}\operatorname{Ind}_T^K \chi_k$. Specialise to $k = 0$, which gives $L^2(\mathrm{SU}(2)/T)$.

(c) Let $\phi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be the covering homomorphism of Lecture 12, with kernel $\{\pm 1\}$. By the display before the weight basis (12.4), $\phi(t_z) = R_3(-\theta)$ for $z = e^{i\theta/2}$, so $\phi(T) = H = \{R_3(\theta)\}$. Show that $F \mapsto F \circ \phi$ is a bijection $C(\mathrm{SO}(3))^H \to C(\mathrm{SU}(2))^T$. Compare (b) with the sphere theorem of Lecture 12.

Exercise 12.3 (Reciprocity maps on SO(3)). Let $K = \mathrm{SO}(3)$, let $H = \{R_3(\theta)\}$ with

$$R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and let $V = \mathbb{C}^3$ with the standard action and inner product. Let $\chi_n(R_3(\theta)) = e^{in\theta}$ on $\mathbb{C}$.

(a) Show that e_3, $e_1 - ie_2$ and $e_1 + ie_2$ are weight vectors. In other words, show that for all θ, $R_3(\theta)$ multiplies them by 1, $e^{i\theta}$ and $e^{-i\theta}$, respectively.

(b) For $n \in \{-1, 0, 1\}$ find a nonzero $A_n \in \operatorname{Hom}_H(V, \chi_n)$. Show that $\operatorname{Hom}_H(V, \chi_n) = 0$ for $|n| \geq 2$.

(c) Write $x \in K$ with columns x_1, x_2, x_3. Compute $F_v = \Psi(A_n)v$ for $n = 0, 1$. Verify the covariance $F_v(xR_3(\theta)) = e^{-in\theta}F_v(x)$ for $n = 1$, and verify $F_{gv} = L_g F_v$.

(d) For which ℓ does V_ℓ occur in $L^2\text{-}\operatorname{Ind}_H^K \chi_n$? Use the weight basis (12.4).

Exercise 12.4 (Homogeneous polynomials on the sphere). Keep the notation of Lecture 12, Section 12.4. Let

$$M(k) = \dim\{f \in \mathcal{P}_\ell : L_{R_3(\theta)} f = e^{ik\theta} f \text{ for all } \theta \in \mathbb{R}\},$$

the dimension of the weight-k space of $\mathcal{P}_\ell$.

(a) Show that $\dim \mathcal{P}_\ell = \frac{1}{2}(\ell+1)(\ell+2)$. Show that $M(k) = \lfloor (\ell - |k|)/2 \rfloor + 1$ for $|k| \le \ell$ and $M(k) = 0$ otherwise.

(b) Deduce that $\mathcal{P}_\ell \cong V_\ell \oplus V_{\ell-2} \oplus \cdots$, ending in V_1 or V_0. Check dimensions for $\ell = 2, 3, 4$.

(c) Deduce that $\mathcal{P}_\ell|_{S^2} = \mathcal{Y}_\ell \oplus \mathcal{Y}_{\ell-2} \oplus \cdots$, and that $\mathcal{P}_\ell|_{S^2} = \mathcal{Y}_\ell \oplus \mathcal{P}_{\ell-2}|_{S^2}$ for $\ell \ge 2$.

Exercise 12.5 (Optional extension: the spherical Laplacian). For $G \in C^2(S^2)$, define $\widetilde{G}(v) = G(v/|v|)$ on $\mathbb{R}^3 \setminus \{0\}$. Set

$$\Delta_{S^2} G = \big(\Delta \widetilde{G}\big)\big|_{S^2}, \qquad \Delta = \partial_x^2 + \partial_y^2 + \partial_z^2.$$

This is the geometric Laplacian of the round sphere. You may use this definition as given. Write $r = |v|$.

(a) Show that $\nabla r^a = a\, r^{a-2} v$ and $\Delta r^a = a(a+1)\, r^{a-2}$ on $\mathbb{R}^3 \setminus \{0\}$.

(b) Let $f \in \mathcal{H}_\ell$. Show that $\Delta_{S^2}(f|_{S^2}) = -\ell(\ell+1)\, f|_{S^2}$.

(c) Use Proposition 12.14 on spherical harmonics and the decomposition in Exercise 12.4(c) to prove $\mathcal{H}_\ell|_{S^2} = \mathcal{Y}_\ell$ and $\dim \mathcal{H}_\ell = 2\ell + 1$.